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EPIDEMIC: Epidemiology Educational Code

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Software

- Review
- Repository
- Archive

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Computational epidemiology

The aim of epidemiology is to understand the causes of a disease in order to predict its evolution and then make decisions to contain the epidemic. Thus, epidemiology is of great importance to society.

The study of a disease spreading leads to the generation of large data sets that requires the use of computational methods to generate epidemiological models and analyze results for decision-making (Tameru et al., 2012). In this context, computational epidemiology arises, which over time has become increasingly multidisciplinary (techniques of epidemiology, biology, mathematics, theoretical computer science, machine learning, etc.) and led to the development of new computational methods to understand and control the spread of disease (Marathe & Ramakrishnan, 2013).

Statement of need

Due to the Covid-19 pandemic, several researchers from different Brazilian institutions organized an initiative called *COVID-19: Observatório Fluminense* (COVID-19RJ) (*COVID-19*, 2020), which aims to monitor the pandemic progress in Brazil, make reliable predictions about the short term evolution and provide high-quality educational material about the mathematical modeling and analysis of COVID-19. To support the research carried out in the COVID-19RJ project, the EPIDEMIC was developed, because through this code it is possible to generate monitoring graphics, trends and disease forecasts. The EPIDEMIC is cited in the reports of the COVID-19RJ project.

In the context of the COVID-19 pandemic, it was observed that computational epidemiology gained prominence, attracting the interest of many people from different areas of knowledge. There are excellent codes available for conducting epidemiological simulations, but these are customized for researchers in the area (Abdulrahman, 2020; Adhikari et al., 2020; Dantas et al., 2018; Hladish et al., 2012; Morrison & Cunha Jr, 2020), so members of COVID-19RJ felt the need to organize EPIDEMIC in a pedagogical way, to collaborate in the training of new researchers. Thus, in addition to EPIDEMIC being a research tool, it is an easy-to-use code that provides a detailed tutorial with several examples, facilitating the insertion of new researchers in the field of epidemiology.

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EPIDEMIC code

The EPIDEMIC code is a suite of basic software for epidemiology that is suitable for analyzing indicators of an epidemic evolution as well as to construct basic compartmental models for qualitative and quantitative analysis. It is developed in an easy to use style, with very lean and well-documented codes. The package also includes an instructional tutorial that gives the end user information about the type of analysis and forecasts that can be obtained with the suite, as well as an overview of the limitations of each model available in the code. The EPIDEMIC code is a didactic and intuitive pedagogical tool for audiences interested in mathematical epidemiology. The EPIDEMIC is developed on the free software platform GNU Octave and is available on a website[1], where one can find a GitHub repository link that directs to the suite source code. The EPIDEMIC code is also compatible with MATLAB proprietary software. The fact that EPIDEMIC is an open package, allows the user to better understand the mathematical structure of the model and is accessible to a large audience. The figure 1 shows the EPIDEMIC logo and the three EPIDEMIC modules: modeling, trends and forecasts.

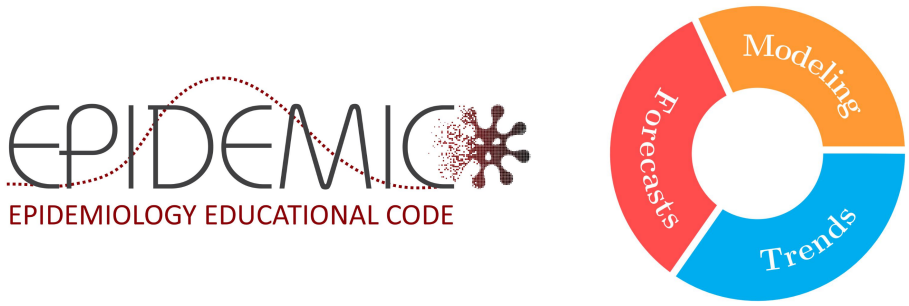

Figure 1: Illustration of the EPIDEMIC code logo and EPIDEMIC modules.

Modeling

In the modeling module, compartmental models, described by differential equations, are used to simulate population dynamics during an epidemic. The compartmental models available are: SIR, SEIR, SIRG, SEIRD, and SEIAHRD.

It is possible with EPIDEMIC to plot the curves of the aforementioned compartmental models and thus analyze the epidemiological dynamics. To do this, just change the values of the parameters of the epidemic that you want to analyze in the code. It is also possible to carry out analyzes, for example, of $\mathcal{R}_0$ in relation to β, or else of the number of deaths in relation to β. In Figure 2 is possible to see an example time series graph of the SEIAHRD compartmental model.

[1] www.EpidemicCode.org

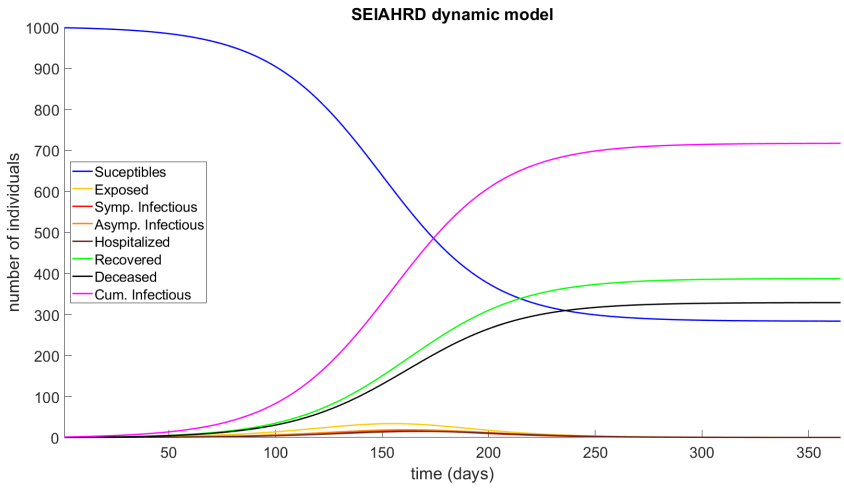

Figure 2: Time series graph of the SEIAHRD dynamic model generated in the modeling module of EPIDEMIC.

Trends

In the trends module, it is possible to monitor, through graphic resources, the behavior of epidemics in countries, states or cities. The analyzes are performed using two basic visualization strategies: that of contagion and mortality and the progress of the epidemic. The types of graphs that can be generated in the trends module are: accumulated deaths, accumulated cases, death progress, case progress, weekly deaths, weekly cases, mortality, prevalence, weekly deaths, and incidence. The Figure 3 shows an example of a trend graph generated in EPIDEMIC. This graph shows the total number of deaths per 1M inhabitants evolution of the COVID-19 pandemic in several countries.

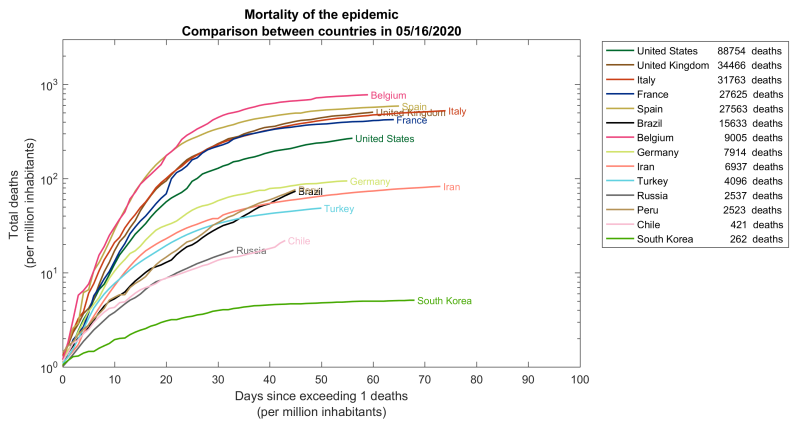

Figure 3: Trend graph of total number of deaths per 1M inhabitants evolution of the COVID-19 pandemic in several countries generated in EPIDEMIC.

Forecasts

In the forecasts module, a statistical regressor is used to obtain forecasts about the short term behavior of the epidemic curves. The method used to obtain the regressor is the classic *Ordinary Least Squares*. In preparing the forecasting code, the following points were taken into account:

- Consider the last five days of the data sample, as these reflect the most recent trend;
- Insert the last five days on the logarithmic scale, as it facilitates visualization in case of exponential growth;
- Plot the predicted values within the estimated reliability envelope. Confidence band is 95%.

The Figure 4 shows a forecast graph of the total cases of COVID-19 in Brazil generated at EPIDEMIC.

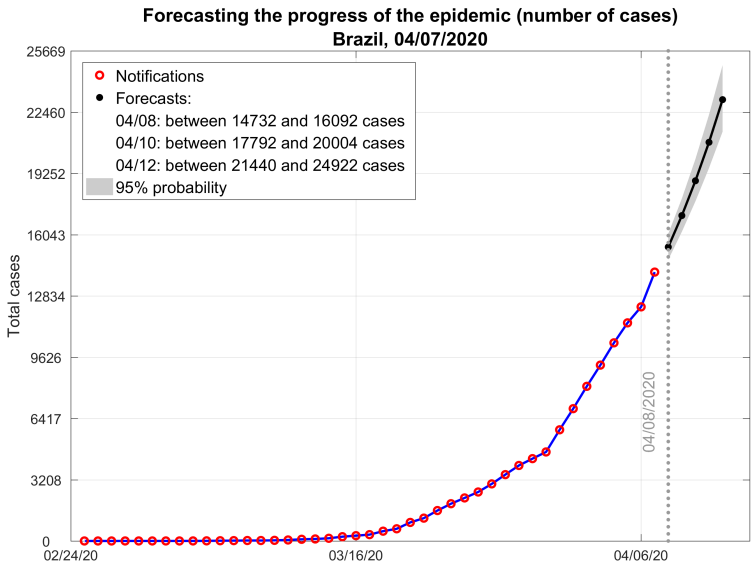

Figure 4: Forecast graph of the total cases of COVID-19 in Brazil generated at EPIDEMIC. The gray color shows the 95% confidence band.

Implementation

For the epidemic implementation, in the modeling module, the package includes the files *main_X.m* and *rhs_X.m*. The "X" in the name represents the compartmental model (SIR, SEIR, SIRG, SEIRD, and SEIAHRD). The file *main* defines the parameters and calculates the reproduction numbers and plots the results of the time series. The File *rhs* defines the ODE system used by the main file. In the trends module, there is the file *epidemic_trends.m*, which is the main file to generate graphs on the numbers of cases and deaths by epidemic in the countries of interest. And in the forecasts module, there is the file *epidemic_forecasts.m*, which is the main file to generate the forecast graphs of accumulated cases and accumulated deaths from an epidemic.

To check the restrictions on the use of EPIDEMIC routines, the "test" folder brings together a set of "verification scripts" to individually cover each possible error in these routines. To facilitate its use, EPIDEMIC has a tutorial, in English and Portuguese, with examples and explanations about the code.

This educational code proves to be an important didactic tool for epidemiological analysis, as it is available in a transparent, accessible and reproducible way (Chatterjee, 2020). Therefore, it is also an important tool for the development of research.

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